



FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
SCHOOL OF ENGINEERING AND ENGINEERING TECHNOLOGY
DEPARTMENT OF COMPUTER ENGINEERING

First (1st) Semester Examination

Course Title: Software Development Techniques (3 Units)

Instruction: Attempt ANY five (5) questions.

2023/2024 Academic Session

Course Code: CPE 407/ICT 303

Time: 2 Hours 15 Minutes

Question 1

- (a) A computer's Central Processing Unit (CPU) cannot directly execute instructions that are written in a language like a spoken language. Explain why this is the case. Give TWO (2) types of software tools that can create machine-code instructions. 4 Marks
- (b) Identify and briefly explain THREE (3) qualities that a good algorithm should have. 6 Marks
- (c) Distinguish between an algorithm, pseudocode, and a flowchart. 6 Marks
- (d) Write an algorithm and draw a flowchart to convert temperature from Celsius to Fahrenheit. 4 Marks

Question 2

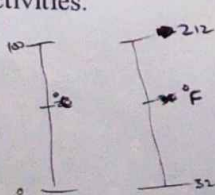
- (a) Describe the Rapid Application Development (RAD) model. 3 Marks
- (b) Differentiate between V-model and Big Bang model. 4 Marks
- (c) With the aid of a diagram, explain the FOUR (4) phases in spiral model. 6 Marks
- (d) Which life cycle model would you follow for developing software for each of the following applications? Justify your selection of model with appropriate reasons.
i. A financial application ii. A new text editor 4 Marks
- (e) How can software crisis be eliminated in the software development process? 3 Marks

Question 3

- (a) What is the difference between Verification and Validation? 3 Marks
- (b) How are requirements validated? 2 Marks
- (c) What is the major distinction between user requirement and system requirement? 2 Marks
- (d) As a software developer hired by a client to build an e-commerce website, list 5 things that are true about the requirements you get from the client. 5 Marks
- (e) What is an SRS document? State and explain 5 characteristics of a good SRS document. 8 Marks

Question 4

- (a) What is Unified Modeling Language (UML)? 2 Marks
- (b) List five main symbols shown on a Class Diagram and explain any two of them in detail. 8 Marks
- (c) What is a use-case diagram? 2 Marks
- (d) The following are the activities of a borrower in a Library System. Construct a use case diagram from these activities.



$$\frac{100-0}{100} = \frac{x-32}{212-32}$$
$$100x = 100x - 32$$
$$20 \rightarrow 68$$
$$\frac{20-0}{100-20} = \frac{x-32}{212-32}$$
$$\frac{20}{80} = \frac{x-32}{180}$$

$$\frac{0-0}{100-0} = \frac{0F-32}{212-32}$$

$$\frac{0C}{100} = \frac{0F-32}{180}$$

$$18^{\circ}C = 10^{\circ}F - 320$$

$$18^{\circ}C + 320 = 10^{\circ}F$$

$$\frac{18^{\circ}C + 32}{10} = ^{\circ}F$$

$$\frac{9}{5}C + 32 = ^{\circ}F$$

$$\frac{9}{5} = 1.8$$

All the best

Borrow Books

Do Research

Reading Newspapers and Periodicals ✓

Get Loan ✓

Check Library Card ✓

Return Books. ✓

8 Marks

Question 5

- (a) ✓ Distinguish between a program and a software product. 3 Marks
- (b) What are the two types of software products? 2 Marks
- (c) What are the advantages and disadvantages of iterative software development model? 4 Marks
- (d) Why does software fail after it has passed from acceptance testing? 3 Marks
- (e) What are the testing principles the software engineer must apply while performing the software testing? 4 Marks
- (f) What are the two levels of testing? List the various testing activities. 4 Marks

Question 6

- (a) ✓ For the scenario described below, which life cycle model would you choose? Give the reason why you would choose this model.
You are interacting with the MIS department of a very large oil company with multiple departments. They have a complex regency system. Migrating the data from this legacy system is not an easy task and would take a considerable time. The oil company is very particular about processes, acceptance criteria and legal contracts. 5 Marks
- (b) What are the principles of agile methods? 5 Marks
- (c) What are the eight (8) levels of software that separates users from hardware? 4 Marks
- (d) Explain the Waterfall model and outline its various phases. 6 Marks

Question 7

- (a) Outline briefly what the following should do in a Scrum team.
i. Product Manager/Owner ii. Software development team iii. Scrum Master 6 Marks
- (b) List and explain the 3 types of software requirements. 7 Marks
- (c) Explain the benefits of using Agile methods like Scrum. 4 Marks
- (d) List three (3) core artifacts in any Scrum framework. 3 Marks